

PASSIVITY UNDER UNMODELED DYNAMICS

H. Shim, J. Byun, J. S. Lee and Jin H. Seo

*School of Electrical Engineering, Seoul National University
San 56-1, Shilim-Dong, Kwanak-Ku, Seoul, Korea, 151-742
Email : hshim@bigfoot.com, Fax : +82-2-8833-251*

Abstract: In this paper, it is pointed out that the passivity can be broken when there is an unmodeled dynamics, and that the phenomenon can be taken place no matter how fast the dynamics is. To prevent the phenomenon, a sufficient condition is also presented which guarantees the preservation of passivity under a sufficiently fast unmodeled dynamics. It is shown that the proposed condition is really met for a physical passive circuit by an example.

Résumé : Dans cet article, on observe que la passivité peut être brisée quand il y a des dynamiques non modélisées et que ce phénomène peut survenir quelle que soit la rapidité des dynamiques. Pour prévenir ce phénomène, on présente une condition suffisante qui garantie que la passivité est préservée sous des dynamiques non modélisées assez rapides. On montre par un exemple que la condition proposée est rencontrée sur un circuit passif.

Keywords: Passive, Singular perturbations, Unmodeled dynamics

1. INTRODUCTION

For modeling a real system, it is usual to consider a system described by the state space representation

$$\dot{x} = F(x) + G(x)u, \quad y = h(x) \quad (1)$$

where a state x is in a set $\mathcal{S}_x \subset R^n$, a control u is in an admissible input set $\mathcal{U} \in R^m$ and an output y is in R^m . Assume that F , G and h are continuously differentiable (C^1), and $F(0) = 0$, $h(0) = 0$.

However, (1) is not usually a whole representation of the real system, since the stable fast (high frequency) dynamics is often neglected in the model to reduce the complexity, and so on. Suppose the full system dynamics as

$$\begin{aligned} \dot{x} &= f_1(x) + Q_1(x)z + g_1(x)u, & y &= h(x) \\ \epsilon \dot{z} &= f_2(x) + Q_2(x)z + g_2(x)u \end{aligned} \quad (2)$$

where $z \in \mathcal{S}_z \subset R^l$ is a state of the fast dynamics and ϵ is a small positive constant which represents two-time scale behavior of the full system

(Kokotovic *et al.*, 1986). Suppose all vector fields and functions are C^1 , and $f_1(0) = 0$, $f_2(0) = 0$. Since most neglected high frequency dynamics in the real world is stable, the following assumption is a reasonable one.

Assumption 1. Assume that there exists a constant $\sigma > 0$ such that

$$\operatorname{Re} \lambda\{Q_2(x)\} \leq -\sigma < 0 \quad (3)$$

that is, the unmodeled dynamics is asymptotically stable for all fixed values of $x \in \mathcal{S}_x$.

By Assumption 1, Q_2 is invertible, and it can be concluded that the reduced system (model system) (1) has a dynamics such that

$$\begin{aligned} F(x) &= f_1(x) - Q_1(x)Q_2^{-1}(x)f_2(x) \\ G(x) &= g_1(x) - Q_1(x)Q_2^{-1}(x)g_2(x). \end{aligned}$$

In the above setting, singular perturbation theory shows that, roughly speaking, “if a reduced

system is asymptotically stable, then the full system is also asymptotically stable for a sufficiently small ϵ under additional technical conditions (Kokotovic *et al.*, 1986; Khalil, 1992; Taylor *et al.*, 1989). In other words, stability is robust under a fast unmodeled dynamics. Now, a question arises naturally, *Is the passivity robust under a fast unmodeled dynamics?*, i.e., when the reduced system (1) is passive, is the full system also passive for a sufficiently small ϵ ? In section 3, it is shown that the answer is negative, by an example. However, it is shown in section 4 that the passivity can be robust by adding some structural conditions on (2). Conclusions are given in section 5.

2. PASSIVE SYSTEMS

In this section, some definitions and lemmas are presented concerning the passivity and passive systems, which is used throughout the paper. Most of materials are adopted from Byrnes *et al.* (1991, Section 2), and some detailed discussions about passivity can be found there, or in other references.

Let a function $V(x)$ be *positive definite* if $V(0) = 0$ and $V(x) > 0$ for $x \neq 0$.

Definition 1. A system is said to be *passive* if there exists a C^0 nonnegative function $V : \mathcal{S}_x \rightarrow R$, called the *storage function*, such that $V(0) = 0$ and, for all $u \in \mathcal{U}$, $x_0 \in \mathcal{S}_x$, $t \geq 0$,

$$V(x(t)) - V(x_0) \leq \int_0^t y^T(\tau)u(\tau) d\tau \quad (4)$$

where $x(t)$ is a solution trajectory of the system with the input u and the initial state x_0 .

If the system is passive with $V(x)$ which is C^r and positive definite function, the system is said to be C^r *PD-passive* for simplicity.

Definition 2. A passive system with storage function V is said to be *strictly passive* if there exists a positive definite function $S : \mathcal{S}_x \rightarrow R$ such that for all $u \in \mathcal{U}$, $x_0 \in \mathcal{S}_x$, $t \geq 0$,

$$V(x(t)) - V(x_0) = \int_0^t y^T(\tau)u(\tau) d\tau - \int_0^t S(x(\tau)) d\tau. \quad (5)$$

Lemma 3. A (strictly) PD-passive system has a Lyapunov (asymptotically) stable zero dynamics.

PROOF. It is easily proved by letting $y \equiv 0$ and differentiating (4) (or, (5)).

For further explanations refer to Byrnes *et al.* (1991), and for definition and discussion about *zero dynamics*, refer to Isidori (1989).

Definition 4. A system is said to have the *strict KYP-property* if there exists a C^1 nonnegative function $V : \mathcal{S}_x \rightarrow R$, with $V(0) = 0$, and a positive definite function $S : \mathcal{S}_x \rightarrow R$, such that

$$\begin{aligned} L_F V(x) &= -S(x) \\ L_G V(x) &= h^T(x) \end{aligned}$$

for each $x \in \mathcal{S}_x$.

Lemma 5. A system which has the (strict) KYP property is (strictly) passive. On the contrary, a (strictly) passive system having a C^1 storage function has the (strict) KYP property.

PROOF. By a slight modification to the proof of Byrnes *et al.* (1991, Proposition 2.12), it can be easily proved.

Throughout the paper, the following notations are used. $\lambda_{\min}(A)$ represents the minimum eigenvalue of matrix A . $|\cdot|$ stands for the Euclidean norm of a vector, or the absolute value of a scalar. $\|A\|$ means the induced matrix norm of A corresponding to the Euclidean norm. $L_f V(x)$ denotes the Lie-derivative of $V(x)$ along the vector field f , i.e. $L_f V(x) = \frac{\partial V}{\partial x} f(x)$.

3. PASSIVITY IS NOT ROBUST UNDER UNMODELED DYNAMICS

In this section, an example is given which, under Assumption 1, shows that passivity is not robust under unmodeled dynamics, i.e., the reduced system (1) is passive (even strictly C^∞ PD-passive), but the full system (2) is not passive.

Example 1.

Suppose the full system as

$$\begin{aligned} \dot{x} &= x + 2z - u, & y &= 9x \\ \epsilon \dot{z} &= -x - z + 5u. \end{aligned} \quad (6)$$

Letting $\epsilon = 0$, the system is reduced to

$$\dot{x} = -x + 9u \quad (7)$$

since the slow manifold is $z = -x + 5u$.

Taking $V(x) = \frac{1}{2}x^2$, (7) is surely passive by Lemma 5. In fact,

$$\begin{aligned} L_F V(x) &= -x^2 \leq 0 \\ L_G V(x) &= 9x = h(x). \end{aligned}$$

However, the zero dynamics of the full system is,

$$\dot{z} = \frac{9}{\epsilon} z \quad (8)$$

which is unstable for all positive ϵ . Since the zero dynamics of the full system is unstable, by Lemma 3, the full system is not PD-passive for any ϵ . (In order to find zero dynamics, at first, let $y \equiv 0$. Thus, $x \equiv 0$ and $u^* = 2z$. This results in (8) from (6).)

Moreover, the full system (6) has a transfer function from u to y ,

$$G(s) = -\frac{9(s - \frac{9}{\epsilon})}{s^2 + (\frac{1}{\epsilon} - 1)s + \frac{1}{\epsilon}}$$

which is not a positive real function, hence, the full system is not passive. (Further discussion about the relation between positive realness and passivity can also be found at Byrnes *et al.* (1991, Prop. 2.10, Remark 2.9).)

4. PASSIVITY CAN BE ROBUST UNDER FAST UNMODELED DYNAMICS

In the previous section, it has been shown the passivity can be broken when there is an unmodeled dynamics even though the dynamics is sufficiently fast. One of the reasons that the full system is not passive seems that $g_2 \neq 0$, i.e., there is a direct path from input to the fast dynamics. For a system to be passive, the passivity inequality (4) should be satisfied for all available input u which may take large value, or may have high frequency. Hence, a path from input to the fast dynamics may break the two-time scale assumptions, i.e., the model reductions are no longer valid since the effects of the unmodeled dynamics are not negligible.

4.1 Linear System Case

Consider the linear system as

$$\begin{aligned} \dot{x} &= F_1 x + Q_1 z + G_1 u, & y &= Hx \\ \epsilon \dot{z} &= F_2 x + Q_2 z + G_2 u \end{aligned} \quad (9)$$

where Q_2 is Hurwitz from Assumption 1, and all matrices have appropriate dimensions.

The system (9) is reduced to

$$\dot{x} = Fx + Gu, \quad y = Hx \quad (10)$$

where $F = F_1 - Q_1 Q_2^{-1} F_2$ and $G = G_1 - Q_1 Q_2^{-1} G_2$.

To guarantee the passivity of the full system, the following assumption seems to be plausible.

Assumption 2. $G_2 \equiv 0$ and $F_2 G_1 \equiv 0$.

Remark 6. The above assumption can be interpreted as follows. In view of (9), the first one eliminates the direct path from input to fast dynamics.

The second one also eliminates the path from the derivative of input to fast dynamics since, in case of $G_2 \equiv 0$, $F_2 x$ is the only path from u to $\dot{z}$ through x . The second assumption can also be seen that the relative degree (Isidori, 1989) should be greater than one when $F_2 x$ is taken as an output of the full system.

The following theorem says that the passivity property is preserved for a sufficiently fast unmodeled dynamics when $G_2 \equiv 0$, $F_2 G_1 \equiv 0$ and Q_2 is Hurwitz.

Theorem 7. Under Assumption 1 and 2, suppose the reduced system (10) is strictly PD-passive, then there exists a positive constant ϵ^* such that for each $\epsilon \in (0, \epsilon^*]$ the full system (9) is also strictly PD-passive.

PROOF. Since the linear system (10) is strictly PD-passive, there exists $V(x) = \frac{1}{2} x^T P x$ which satisfies (5), where $P > 0$ and $P^T = P$. By Lemma 5 and Assumption 2,

$$\begin{aligned} PF + F^T P &= -\Gamma \\ PG &= PG_1 = H^T \end{aligned}$$

where Γ is a positive definite matrix.

To describe the behavior more easily, the full system (9) can be transformed to the followings by a change of variable $\eta = z + Q_2^{-1} F_2 x$, using the fact $F_2 G_1 \equiv 0$,

$$\begin{aligned} \dot{x} &= Fx + Gu + Q_1 \eta, \\ \epsilon \dot{\eta} &= Q_2 \eta \\ &\quad + \epsilon (Q_2^{-1} F_2 F x + Q_2^{-1} F_2 G u + Q_2^{-1} F_2 Q_1 \eta) \\ &= Q_2 \eta + \epsilon (Q_2^{-1} F_2 F x + Q_2^{-1} F_2 Q_1 \eta) \\ y &= Hx. \end{aligned} \quad (11)$$

Consider now the positive definite function

$$W(x, \eta) = V(x) + \frac{1}{2} \eta^T P_\eta \eta = \frac{1}{2} x^T P x + \frac{1}{2} \eta^T P_\eta \eta$$

where P_η is such that $P_\eta Q_2 + Q_2^T P_\eta = -I$. Such P_η is well-defined since Q_2 is Hurwitz. Regarding the system (11) as

$$\begin{pmatrix} \dot{x} \\ \dot{\eta} \end{pmatrix} = \bar{f}(x, \eta) + \bar{g}(x, \eta) u$$

where

$$\begin{aligned} \bar{f}(x, \eta) &= \begin{pmatrix} Fx + Q_1 \eta \\ \frac{1}{\epsilon} Q_2 \eta + Q_2^{-1} F_2 F x + Q_2^{-1} F_2 Q_1 \eta \end{pmatrix} \\ \bar{g}(x, \eta) &= \begin{pmatrix} G \\ 0 \end{pmatrix}. \end{aligned}$$

A straightforward calculation shows that

$$\begin{aligned}
L_{\bar{f}}W(x, \eta) + L_{\bar{g}}W(x, \eta)u &= \dot{W}(x, \eta) \\
&= -\frac{1}{2}x^T \Gamma x + x^T P G u + x^T P Q_1 \eta \\
&\quad - \frac{1}{2\epsilon} \eta^T \eta + \eta^T P_{\eta} Q_2^{-1} F_2 F x \\
&\quad + \eta^T P_{\eta} Q_2^{-1} F_2 Q_1 \eta \\
&= -\frac{1}{2} \begin{pmatrix} x \\ \eta \end{pmatrix}^T \begin{pmatrix} \Gamma & -N^T \\ -N & \frac{1}{\epsilon} I - M \end{pmatrix} \begin{pmatrix} x \\ \eta \end{pmatrix} \\
&\quad + x^T P G u
\end{aligned}$$

where

$$\begin{aligned}
M &= (P_{\eta} Q_2^{-1} F_2 Q_1) + (P_{\eta} Q_2^{-1} F_2 Q_1)^T \\
N &= Q_1^T P + P_{\eta} Q_2^{-1} F_2 F.
\end{aligned}$$

For M , N and Γ are independent of ϵ , there exists a constant ϵ^* such that for each $\epsilon \in (0, \epsilon^*)$

$$\begin{pmatrix} \Gamma & -N^T \\ -N & \frac{1}{\epsilon} I - M \end{pmatrix}$$

is positive definite. For such small ϵ , there exists a positive definite function $S(x, \eta)$ such that,

$$L_{\bar{f}}W(x, \eta) + L_{\bar{g}}W(x, \eta)u = -S(x, \eta) + y^T u \quad (12)$$

using the fact that $L_{\bar{g}}W(x, \eta) = L_G V(x) = Hx = y$ by Assumption 2.

Clearly, integrating both sides of (12) yields (5). Thus, the full system is also strictly PD-passive with $W(x, \eta) = W(x, z - Q_2^{-1} F_2 x)$. This completes the proof.

The assumption 2 may seem to be restrictive. However, the following example shows that a physical passive circuit does satisfy Assumption 2. Since it can be conjectured that a circuit, made of passive elements such as resistors, capacitors and inductors, is surely passive, the full order model and reduced order model would be both passive. Thus, the example shows that a physical system, which is passive in a full order form and a reduced order form, satisfies the assumption 2.

Example 2.

This example is adopted from Vidyasagar (1993, p.133). Consider the circuit shown in Figure 1. In this circuit, the clear choice of state variables is x_1 , x_2 and z , where x_1 is the current of the inductor, x_2 and z are the voltages of the capacitor C_1 and C_p respectively. When C_p is sufficiently small, it can be regarded as a parasitic capacitor, thus, it is neglected and can be viewed as a disconnected line as Figure 2.

The standard circuit theory leads to the following system equation from Figure 1,

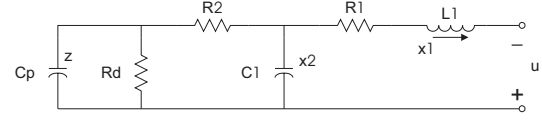


Fig. 1. Full order model

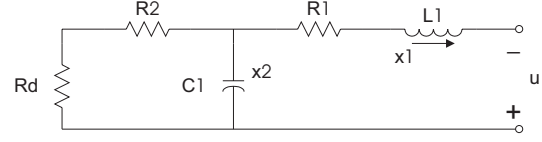


Fig. 2. Reduced order model

$$\begin{aligned}
\dot{x} &= \begin{bmatrix} -\frac{R_1}{L_1} & \frac{1}{L_1} \\ -\frac{1}{C_1} & -\frac{1}{C_1 R_2} \end{bmatrix} x + \begin{bmatrix} 0 \\ \frac{1}{C_1 R_2} \end{bmatrix} z + \begin{bmatrix} \frac{1}{L_1} \\ 0 \end{bmatrix} u \\
\epsilon \dot{z} &= \begin{bmatrix} 0 & \frac{1}{R_2} \end{bmatrix} x - \frac{R_d + R_2}{R_d R_2} z + 0 u \\
y &= x_1
\end{aligned} \quad (13)$$

where $\epsilon = C_p$, which is assumed to be small. And the input and the output are assumed to be a driving voltage u , and the current x_1 respectively.

Similarly, from Figure 2, the following reduced system equation is deduced.

$$\begin{aligned}
\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} &= \begin{bmatrix} -\frac{R_1}{L_1} & \frac{1}{L_1} \\ -\frac{1}{C_1} & -\frac{1}{C_1(R_d + R_2)} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} \frac{1}{L_1} \\ 0 \end{bmatrix} u \\
y &= x_1
\end{aligned} \quad (14)$$

which is also obtained by letting $\epsilon = 0$ in (13).

It is easily checked that the reduced system (14) is strictly PD-passive. Indeed, a storage function $V(x) = \frac{1}{2}x^T P x$, in which

$$P = \begin{bmatrix} L_1 & 0 \\ 0 & C_1 \end{bmatrix},$$

satisfies the Lemma 5.

On the other hand, Assumption 2 is clearly satisfied in (13). Hence the full system (13) is also strictly PD-passive for small ϵ . These conclusions coincide with the previous conjectures. $\square$

4.2 Nonlinear System Case

The result of previous subsection is extended to nonlinear case. As in the linear case, consider the following condition.

Assumption 3. In the representation of the full system (2), assume that

- (1) $g_2(x) \equiv 0$
- (2) $L_G Q_2^{-1} F_2(x) = \frac{\partial}{\partial x} (Q_2^{-1}(x) F_2(x)) G(x) \equiv 0$

for all $x \in \mathcal{S}_x$.

Theorem 8. Suppose the nonlinear reduced system (1) is strictly C^1 PD-passive on $\mathcal{S}_x$, in other

words, there exist a C^1 storage function $V(x)$ and a positive definite function $S(x)$ on $\mathcal{S}_x$ such that

$$L_F V(x) = -S(x), \quad L_G V(x) = -h^T(x). \quad (15)$$

Moreover, assume that there exist c ($c > 0$) and ρ ($0 < \rho \leq 1$) such that

$$S(x) \geq c |x|^{2\rho}, \quad \forall x \in \mathcal{S}_x. \quad (16)$$

Then, under Assumption 1 and 3, there exists a connected set $\Omega_x(\subset \mathcal{S}_x)$ and a positive constant ϵ^* such that, for each $\epsilon \in (0, \epsilon^*]$, the full nonlinear system (2) is also strictly C^1 PD-passive on $\Omega_x \times \mathcal{S}_z$.

PROOF. By a change of coordinates $\eta = z + Q_2^{-1}(x)f_2(x)$, the full system (2) becomes

$$\begin{aligned} \dot{x} &= F(x) + G(x)u + Q_1(x)\eta \\ \epsilon \dot{\eta} &= Q_2(x)\eta + \\ &\quad \epsilon \left(\frac{\partial(Q_2^{-1}f_2)}{\partial x} (F(x) + G(x)u + Q_1(x)\eta) \right) \\ &= Q_2(x)\eta + \\ &\quad \epsilon \left(\frac{\partial(Q_2^{-1}f_2)}{\partial x} F(x) + \frac{\partial(Q_2^{-1}f_2)}{\partial x} Q_1(x)\eta \right) \\ y &= h(x) \end{aligned} \quad (17)$$

under the Assumption 3.

Since $Q_2(0)$ is Hurwitz from Assumption 1, there exists the positive definite symmetric matrix P_η such that

$$P_\eta Q_2(0) + Q_2^T(0)P_\eta = -I.$$

It follows that, by the continuity of $Q_2(x)$, there exists a compact set $\Omega_x \subset \mathcal{S}_x$ such that

$$\Gamma_\eta(x) := -(P_\eta Q_2(x) + Q_2^T(x)P_\eta) \quad (18)$$

is positive definite for each $x \in \Omega_x$.

Now, consider a positive definite storage function for the transformed full system (17) as

$$W(x, \eta) = V(x) + \frac{1}{2}\eta^T P_\eta \eta,$$

and rewrite the system (17) as

$$\begin{pmatrix} \dot{x} \\ \dot{\eta} \end{pmatrix} = \bar{f}(x, \eta) + \bar{g}(x, \eta)u$$

where

$$\begin{aligned} \bar{f}(x, \eta) &= \begin{bmatrix} F(x) + Q_1(x)\eta \\ \frac{1}{\epsilon} Q_2(x)\eta + \frac{\partial(Q_2^{-1}f_2)}{\partial x} F(x) + \frac{\partial(Q_2^{-1}f_2)}{\partial x} Q_1(x)\eta \end{bmatrix} \\ \bar{g}(x, \eta) &= \begin{bmatrix} G(x) \\ 0 \end{bmatrix}. \end{aligned}$$

Before evaluating the derivative of W , define $M(x)$ and $N'(x)$ for the notational convenience.

$$P_\eta \frac{\partial(Q_2^{-1}f_2)}{\partial x}(x) Q_1(x) =: M(x) \quad (19)$$

$$(L_{Q_1} V)^T(x) + P_\eta \frac{\partial(Q_2^{-1}f_2)}{\partial x}(x) F(x) =: N'(x)x. \quad (20)$$

Existence of such $N'(x)$ is followed from the facts that $F(0) = 0$ and that $\frac{\partial V}{\partial x}(0) = 0$ ($V(x)$ has the minimum at $x = 0$) (Nijmeijer and Van der Schaft, 1990, p.39). From (19) and (20), $M(x)$ and $N'(x)$ are continuous on $\mathcal{S}_x$.

Hence,

$$\begin{aligned} L_{\bar{f}} W(x, \eta) + L_{\bar{g}} W(x, \eta)u &= \dot{W}(x, \eta) \quad (21) \\ &= L_F V + L_G V u + L_{Q_1} V \eta - \frac{1}{2\epsilon} \eta^T \Gamma(x) \eta + \\ &\quad \eta^T P_\eta \left(\frac{\partial(Q_2^{-1}f_2)}{\partial x} F(x) + \frac{\partial(Q_2^{-1}f_2)}{\partial x} Q_1(x)\eta \right) \\ &= L_G V u + L_F V + \eta^T N'(x)x \\ &\quad - \frac{1}{2\epsilon} \eta^T \Gamma(x) \eta + \eta^T M(x) \eta \\ &\leq L_G V u - c |x|^{2\rho} + |\eta| \|N'(x)\| |x| \\ &\quad - \frac{\lambda_{\min}(\Gamma(x))}{2\epsilon} |\eta|^2 + \|M(x)\| |\eta|^2 \end{aligned}$$

by defining $N(x) \equiv N'(x)|x|^{1-\rho}$,

$$\begin{aligned} &= L_G V u - c |x|^{2\rho} + |\eta| \|N(x)\| |x|^\rho \\ &\quad - \frac{\lambda_{\min}(\Gamma(x))}{2\epsilon} |\eta|^2 + \|M(x)\| |\eta|^2 \\ &= L_G V u - \frac{1}{2} \left(\frac{|x|^\rho}{|\eta|} \right)^T \Phi \left(\frac{|x|^\rho}{|\eta|} \right) \quad (22) \end{aligned}$$

where

$$\Phi = \begin{bmatrix} 2c & -\|N(x)\| \\ -\|N(x)\| & \frac{1}{\epsilon} \lambda_{\min}(\Gamma(x)) - 2\|M(x)\| \end{bmatrix}.$$

Here, $N(x)$ and $M(x)$ are bounded on Ω_x since they are continuous on a compact set Ω_x , and $\Gamma(x)$ is positive definite on Ω_x by (18). Thus, it follows that there exists a positive constant ϵ^* such that for each $\epsilon \in (0, \epsilon^*]$ the matrix Φ is negative definite. Indeed,

$$\frac{2}{\epsilon^*} \lambda_{\min}(\Gamma(x)) - 4\|M(x)\| - \|N(x)\|^2 > 0 \quad \text{on } B_r.$$

On the other hands, $y = h(x) = (L_{\bar{g}} W)^T = (L_G V)^T$.

By comparing (21) and (22), the strict KYP property holds for the full system, which, by Lemma 5, implies the full system is strictly C^1 PD-passive on $\Omega_x \times \mathcal{S}_z$. $\square$

5. CONCLUSIONS

Recently, passification method for a non-passive system has been developed, and several control problems under the passive property have been studied, e.g. Byrnes *et al.* (1991), Su and Xie (1996) and Jiang *et al.* (1996). This paper gives an early warning by the fact that, even if a system is made passive on the order reduced model, it may not be valid in real situations. Especially, it is shown that the passivity can be broken under an unmodeled dynamics. In the example of section

3, it occurs even though the unmodeled dynamics are sufficiently fast.

A sufficient structural conditions are also presented with which the system under a sufficiently fast unmodeled dynamics remains passive when the reduced system is passive. It is analyzed for linear and nonlinear systems separately, and it is shown that, by an example, the proposed structural conditions are really satisfied for the physical passive circuit.

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